

Vectors in C++ - Practice Questions & Solutions

This document includes 10 solved programs demonstrating key vector operations in C++. Each example shows proper use of STL functions, loops, and common manipulations like insertion, deletion, sorting, and merging.

Q1. Create and Display a Vector

```
#include <iostream>
#include <vector>
using namespace std;
int main() {
    vector<int> v; int n, x;
    cout << "Enter number of elements: "; cin >> n;
    cout << "Enter elements:\n";
    for (int i = 0; i < n; i++) { cin >> x; v.push_back(x); }
    cout << "Vector elements: ";
    for (int i = 0; i < v.size(); i++) cout << v[i] << " ";
}
```

Q2. Sum and Average of Elements

```
#include <iostream>
#include <vector>
using namespace std;
int main() {
    int n, x, sum = 0; vector<int> v;
    cout << "Enter number of elements: "; cin >> n;
    for (int i = 0; i < n; i++) { cin >> x; v.push_back(x); }
    for (int val : v) sum += val;
    cout << "Sum = " << sum << endl;
    cout << "Average = " << (float)sum / v.size();
}
```

Q3. Find Maximum and Minimum Element

```
#include <iostream>
#include <vector>
#include <algorithm>
using namespace std;
int main() {
    int n, x; vector<int> v;
    cout << "Enter number of elements: "; cin >> n;
    for (int i = 0; i < n; i++) { cin >> x; v.push_back(x); }
    cout << "Max = " << *max_element(v.begin(), v.end()) << endl;
    cout << "Min = " << *min_element(v.begin(), v.end()) << endl;
}
```

Q4. Delete Last Element from Vector

```
#include <iostream>
#include <vector>
using namespace std;
int main() {
    vector<int> v = {10, 20, 30, 40, 50};
    cout << "Before pop_back: "; for (int x : v) cout << x << " ";
    v.pop_back();
    cout << "\nAfter pop_back: "; for (int x : v) cout << x << " ";
}
```

Q5. Insert Element at Specific Position

```
#include <iostream>
#include <vector>
using namespace std;
int main() {
    vector<int> v = {10, 20, 30, 40, 50};
    int pos, val; cout << "Enter position and value: "; cin >> pos >> val;
    if (pos >= 0 && pos <= v.size()) v.insert(v.begin() + pos, val);
    cout << "Updated vector: "; for (int x : v) cout << x << " ";
}
```

Q6. Reverse Elements of Vector

```
#include <iostream>
#include <vector>
#include <algorithm>
using namespace std;
int main() {
    vector<int> v = {1, 2, 3, 4, 5};
    reverse(v.begin(), v.end());
    cout << "Reversed vector: "; for (int x : v) cout << x << " ";
}
```

Q7. Sort a Vector

```
#include <iostream>
#include <vector>
#include <algorithm>
using namespace std;
int main() {
    vector<int> v = {5, 2, 9, 1, 3};
    sort(v.begin(), v.end());
    cout << "Ascending: "; for (int x : v) cout << x << " ";
    sort(v.rbegin(), v.rend());
    cout << "\nDescending: "; for (int x : v) cout << x << " ";
}
```

Q8. Remove Duplicate Elements

```
#include <iostream>
#include <vector>
#include <algorithm>
using namespace std;
int main() {
    vector<int> v = {4, 2, 2, 4, 3, 3, 1};
    sort(v.begin(), v.end());
    v.erase(unique(v.begin(), v.end()), v.end());
    cout << "After removing duplicates: "; for (int x : v) cout << x << " ";
}
```

Q9. Merge Two Vectors

```
#include <iostream>
#include <vector>
using namespace std;
int main() {
    vector<int> v1 = {1, 2, 3};
    vector<int> v2 = {4, 5, 6};
    vector<int> v3;
    v3.insert(v3.end(), v1.begin(), v1.end());
    v3.insert(v3.end(), v2.begin(), v2.end());
    cout << "Merged vector: "; for (int x : v3) cout << x << " ";
}
```

Q10. 2D Vector Input and Display

```
#include <iostream>
#include <vector>
using namespace std;
int main() {
    int m, n; cout << "Enter rows and columns: "; cin >> m >> n;
    vector<vector<int>> mat(m, vector<int>(n));
    cout << "Enter elements:\n";
    for (int i = 0; i < m; i++)
        for (int j = 0; j < n; j++)
            cin >> mat[i][j];
    cout << "Matrix:\n";
    for (int i = 0; i < m; i++) {
        for (int j = 0; j < n; j++)
            cout << mat[i][j] << " ";
        cout << endl;
    }
}
```